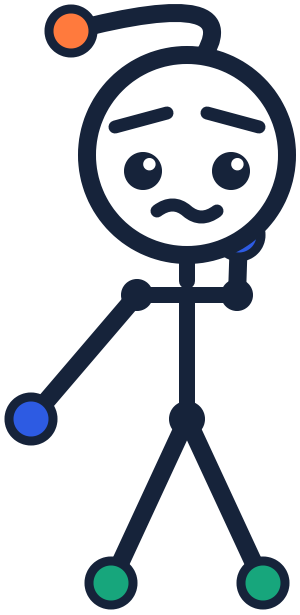
Young Innovators
FOR CHANGE
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A FULL-YEAR ENRICHMENT PROGRAM · GRADE 4

# It Cannot Be Done

36 weekly assignments on proving things, including the things that *cannot happen*



**This book belongs to**

_____

Teacher _____ School year _____

One assignment a week · two pages, front and back · about 30 to 40 minutes
Built to sit alongside the California Grade 4 math curriculum, not to replace it.

# How This Book Works

Read this page with a grown-up before Week 1.



**Hello again. I am Pip.**

Grade 3 was about counting things properly. This year is about a harder move: showing that something **cannot happen at all**, without trying every case. In Week 11 you will settle a puzzle that beat a whole city for years, in three lines.

### One assignment a week, two pages

**Page 1** has a note about what you are doing in class this week, a two-minute **Number Warm-Up**, the **Words to keep**, and the main problem.

**Page 2** has more to try, then **Add to your Proof File**, a **Talk About It** question, a hint, and a ★ **Challenge Zone**.

### The Proof File

One folder, built all year, one page per proof. Each page has a **claim** and a **reason**, never just an answer. By Week 36 it is about thirty-three pages and it is the part you keep. Get the folder in Week 1.

### Three rules for this book

1. **I tried and could not** is not a proof. It is a report on your afternoon.
2. To knock a claim down you need **one** example. To hold it up you need a reason that covers every case.
3. When a reason works, stop looking for more examples. That is what a reason is for.

### For the grown-up helping

The question worth repeating all year is **“how do you know that covers every case?”** Many children arrive in Grade 4 believing that enough examples settle a question. Most of this book is the slow correction of that belief, and it takes the whole year, so do not rush a page where the answer is already obvious to them. The answer is never the point here. The reason is.

# The Year at a Glance

Weeks 1–18. The last column is what the class is usually doing that week.

| WK | ASSIGNMENT | STRAND | IN CLASS THIS WEEK |
|---|---|---|---|
| **TRIMESTER 1 · WEEKS 1–12** | | | |
| 1 | **The Thing That Never Changes** | Logic & Proof | Number and shape patterns (4.OA.5) |
| 2 | **Odd, Even and Impossible** | Number Structure | Number patterns; place value (4.OA.5, 4.NBT.4) |
| 3 | **Factor Rectangles** | Number Structure | Factors, multiples, prime and composite (4.OA.4) |
| 4 | **Primes and Composites** | Number Structure | Prime and composite numbers (4.OA.4) |
| 5 | **Building From Primes** | Number Structure | Factors and multiples; multi-digit multiplication (4.OA.4, 4.NBT.5) |
| 6 | **Multiples That Meet** | Number Structure | Factors and multiples (4.OA.4) |
| 7 | **The Biggest Shared Factor** | Number Structure | Factors and multiples (4.OA.4) |
| 8 | **Coloring Proves It** | Logic & Proof | Multi-step problems and reasoning (4.OA.3) |
| 9 | **The Domino Problem** | Logic & Proof | Multi-step problems and reasoning (4.OA.3) |
| 10 | **Degree Tells You** | Dots & Lines | Multi-step problems and reasoning (4.OA.3) |
| 11 | **Trace It or Prove You Cannot** | Dots & Lines | Multi-step problems and reasoning (4.OA.3) |
| 12 | **Impossible Fair** | Logic & Proof | Multi-step problems and reasoning (4.OA.3) |
| **TRIMESTER 2 · WEEKS 13–24** | | | |
| 13 | **Case Files** | Logic & Proof | Review and reasoning |
| 14 | **Divisible by 2, 5 and 10** | Number Structure | Place value to one million (4.NBT.1, 4.NBT.2) |
| 15 | **Divisible by 3 and 9** | Number Structure | Place value and multi-digit arithmetic (4.NBT.1, 4.NBT.4) |
| 16 | **Casting Out Nines** | Number Structure | Multi-digit multiplication (4.NBT.5) |
| 17 | **Wrapping Around, Again** | Number Structure | Multi-digit arithmetic and patterns (4.NBT.4, 4.OA.5) |
| 18 | **The Day of the Week** | Number Structure | Multi-step problems; measurement (4.OA.3, 4.MD.2) |

# The Year at a Glance

Weeks 19–36. The last column is what the class is usually doing that week.

| WK | ASSIGNMENT | STRAND | IN CLASS THIS WEEK |
|---|---|---|---|
| 19 | **Lines of Symmetry** | Shape & Space | Lines of symmetry in two-dimensional figures (4.G.3) |
| 20 | **Turning a Shape** | Shape & Space | Lines of symmetry; angle measure (4.G.3, 4.MD.5) |
| 21 | **The Eight Ways to Move a Square** | Shape & Space | Angle measure and reasoning (4.MD.5, 4.OA.3) |
| 22 | **Angles That Must Add Up** | Shape & Space | Angle measure and adding angles (4.MD.5, 4.MD.7) |
| 23 | **Tiling the Floor** | Shape & Space | Angle measure (4.MD.5, 4.MD.7) |
| 24 | **Impossible Tilings** | Shape & Space | Multi-step problems and reasoning (4.OA.3) |
| **TRIMESTER 3 · WEEKS 25–36** | | | |
| 25 | **Puzzle Fair** | Logic & Proof | Review and reasoning |
| 26 | **Choosing a Team** | Ways of Counting | Multi-step problems (4.OA.3) |
| 27 | **The Triangle Grows** | Ways of Counting | Number and shape patterns (4.OA.5) |
| 28 | **The Triangle and the Teams** | Ways of Counting | Number and shape patterns (4.OA.5) |
| 29 | **Trees** | Dots & Lines | Multi-step problems and reasoning (4.OA.3) |
| 30 | **The Cheapest Network** | Dots & Lines | Multi-step problems (4.OA.3) |
| 31 | **Two Ways to Multiply** | Steps & Rules | Multiplying multi-digit numbers (4.NBT.5) |
| 32 | **The Euclid Machine** | Steps & Rules | Multi-step problems; factors (4.OA.3, 4.OA.4) |
| 33 | **The Same Point on the Line** | Fractions | Equivalent fractions (4.NF.1, 4.NF.2) |
| 34 | **Parts of a Hundred** | Fractions | Decimal notation for fractions (4.NF.5, 4.NF.6) |
| 35 | **The Weighing Puzzles** | Logic & Proof | Multi-step problems and reasoning (4.OA.3) |
| 36 | **The Proof File, Finished** | Logic & Proof | Review of the year |

WEEK 1 LOGIC & PROOF

# The Thing That Never Changes

Find what a move cannot touch, and you can rule out the future.

**IN CLASS THIS WEEK**
Number and shape patterns (4.OA.5)

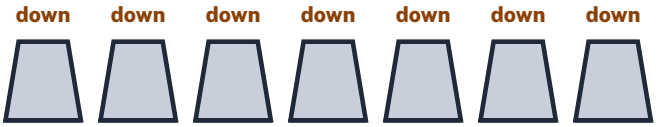
**NUMBER WARM-UP · 2 MINUTES**
Is 27 odd or even? Add 2. Still odd? Add 2 again.

**WORDS TO KEEP**

**move** one action the rules allow you to take

**invariant** something that stays the same no matter which moves you make, or how many

Seven cups, all upside down. The one rule: **turn over exactly two cups**, any two you like. Your goal is to get every cup the right way up.




the starting position

## 1 Watch one number.

a. How many cups are upside down at the start? _____

b. A move turns two cups. There are three kinds. Write what each one does to the count: does it go up by 2, down by 2, or stay the same?

two down cups turned up: _____

one down and one up: _____

two up cups turned down: _____

c. All three changes are _____ numbers, so the count can never flip between odd and even.

## 2 Now the goal.

a. To finish, how many cups must be upside down? _____ Is that odd or even? _____

b. You started odd, and every move keeps it odd. So can you ever finish? _____

c. How many moves did you have to try to be sure? _____

## 3 Six cups instead.

Same rule, but now **six** cups start upside down.

a. Is it possible now? _____

b. Show it. Write which two cups you turn on each move, and how many moves you need.

_____

_____

**✎ ADD TO YOUR PROOF FILE**

Start the Proof File. Page 1 is this claim, written out in three lines: what you cannot do, the number that never changes, and why that settles it.

**💬 TALK ABOUT IT**

You proved something about moves nobody has made yet. How is that different from trying for an hour and giving up?

The invariant here is not the number of upside-down cups. It is something about that number. Say exactly what.



**STUCK? TRY THIS**

Do not look for a clever sequence of moves. Look for a number that every single move leaves alone, and then check whether the goal has the right value of it.

**★ CHALLENGE ZONE**

Same seven cups, but now you must turn over exactly **three** at a time. Is it possible? Try it before you decide, because the answer is not the same.

Explain why the answer changed when the number 2 changed to 3.

WEEK 2 NUMBER STRUCTURE

# Odd, Even and Impossible

Odd and even is not a label. It is a tool that rules things out.

**IN CLASS THIS WEEK**
Number patterns; place value (4.OA.5, 4.NBT.4)

**NUMBER WARM-UP · 2 MINUTES**
odd + odd = ? odd + even = ? even + even = ?

**WORDS TO KEEP**
**parity** whether a number is odd or even, thought of as a property that survives adding and subtracting

Fill this in from the warm-up. It is the whole toolkit for the week.

| | **+ even** | **+ odd** |
|---|---|---|
| even | | |
| odd | | |

## 1 Add a lot of them.

a. Add **four** even numbers. Odd or even? _____

b. Add **four** odd numbers. _____

c. Add **five** odd numbers. _____

d. Write the rule for adding **n** odd numbers.

_____

## 2 Now rule things out.

| Question | possible? | the reason |
|---|---|---|
| 15 as the sum of 4 even numbers | | |
| 21 as the sum of 6 odd numbers | | |
| 21 as the sum of 5 odd numbers | | |
| 30 as the sum of 3 odd numbers | | |

## 3 One of them is possible.

Pick the row you marked possible and actually write the numbers out. Ruling something out is not the same as building it.

✎ **ADD TO YOUR PROOF FILE**

Proof File page 2: the parity table, and underneath it the sentence **a sum of even numbers is always even, so anything odd is out of reach.**

💬 **TALK ABOUT IT**

Why can a rule about odd and even settle a question about numbers you never wrote down?

Somebody says 15 is impossible because they tried lots of even numbers and none worked. Is that the same proof you gave?



**STUCK? TRY THIS**

Do not add the actual numbers. Ask only whether the total **has to be** odd or even, then compare that with the target.

★ **CHALLENGE ZONE**

Write the numbers 1 to 10 in a row. Put a + or a − in front of each one, however you like. Can the answer ever be exactly 0?

Work out the total of 1 to 10 first. Then think about what changing a + to a − does to that total.

WEEK 3 NUMBER STRUCTURE

# Factor Rectangles

Every rectangle you can build from a number is a fact about that number.

**IN CLASS THIS WEEK**
Factors, multiples, prime and composite (4.OA.4)

**NUMBER WARM-UP · 2 MINUTES**
What times 6 makes 24? What times 8 makes 24?

**WORDS TO KEEP**

**factor** a number that divides another exactly, with nothing left over

**factor pair** two factors that multiply to make the number

Every rectangle that can be built from exactly **24** squares. You met this idea in Grade 3. This year it is a tool, not a picture.



24 × 1 12 × 2 8 × 3 6 × 4

**1 Read the factors off the rectangles.**

a. Write the four factor pairs of 24.

_____

b. List every factor of 24, smallest first.

_____

c. How many factors does 24 have? _____

## 2 Three more.

| Number | factor pairs | all its factors | how many |
|---|---|---|---|
| 36 | | | |
| 48 | | | |
| 13 | | | |

## 3 The square one.

a. Which of your numbers can be built as a **square**? _____

b. Count its factors. It is the only one with an **odd** count. Say why a square rectangle causes that.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 3: the factor rectangles of 24, and the sentence explaining why a square number has an odd number of factors.

**💬 TALK ABOUT IT**

Why does every factor come in a pair? Where does the pairing break down?

Could a number have exactly three factors? Try to build one before you answer.



**STUCK? TRY THIS**

Divide by 1, then 2, then 3, and so on. Every exact division is a factor pair. Stop when the pairs start repeating themselves.

**★ CHALLENGE ZONE**

Which number below 50 has the most factors? Check every likely candidate rather than guessing.

Find every number below 50 with exactly three factors. They have something in common that next week explains.

WEEK 4 NUMBER STRUCTURE

# Primes and Composites

Cross out every number that is a multiple of something. Look at what survives.

**IN CLASS THIS WEEK**
Prime and composite numbers (4.OA.4)

**NUMBER WARM-UP · 2 MINUTES**
Is 51 prime? Is 53?

**WORDS TO KEEP**

**prime** a number with exactly two factors, 1 and itself

**composite** a number with more than two factors

**the sieve** crossing out multiples until only primes are left

Circle **2**, then cross out all the other multiples of 2. Circle **3**, cross out its other multiples. Then **5**. Then **7**. Stop there.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 | 78 | 79 | 80 |
| 81 | 82 | 83 | 84 | 85 | 86 | 87 | 88 | 89 | 90 |
| 91 | 92 | 93 | 94 | 95 | 96 | 97 | 98 | 99 | 100 |

## 1 What survived.

a. How many primes are there below 100? _____

b. What is the largest one? _____

c. Why was it safe to stop at 7? The next prime is 11, and 11 × 11 = _____, which is bigger than 100.

## 2 Test three numbers properly.

| Number | prime? | if not, a factor pair |
|---|---|---|
| 91 | | |
| 97 | | |
| 87 | | |

## 3 The awkward two.

a. Is 1 prime? Count its factors first. _____

b. Is 2 prime? It is the only even one, which makes people doubt it. _____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 4: your list of primes below 100, and the reason the sieve could stop at 7. That reason is the useful part, not the list.

**💬 TALK ABOUT IT**

The sieve never asks whether a number is prime. It only crosses out. Why does that work?

Are the primes spread out evenly? Count how many are in each row of ten.



**STUCK? TRY THIS**

To test one number, try dividing by 2, 3, 5, 7, 11 and so on, and stop as soon as the divisor times itself passes your number.

**★ CHALLENGE ZONE**

Twin primes are two primes that differ by 2, like 11 and 13. Find every twin pair below 50.

Find the longest run of composite numbers below 100 with no prime in it.

WEEK 5 NUMBER STRUCTURE

# Building From Primes

Break a number apart however you like. The bottom of the tree is always the same.

**IN CLASS THIS WEEK**
Factors and multiples; multi-digit multiplication (4.OA.4, 4.NBT.5)

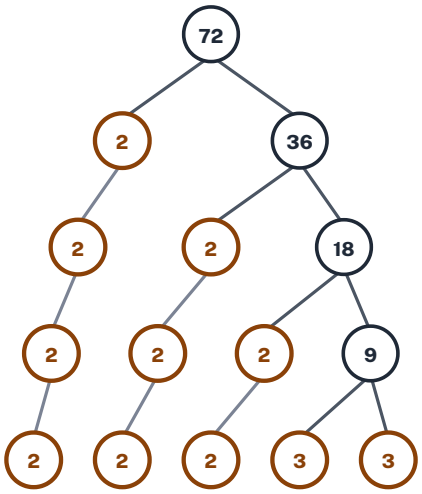
**NUMBER WARM-UP · 2 MINUTES**
What is 8 × 9? Now build 72 a different way.

**WORDS TO KEEP**

**factor tree** a picture of a number broken into factors, then those broken again, until only primes are left

**prime factorization** the list of primes at the bottom of the tree

A factor tree for **72**. The circled numbers are prime, so they never split again. Each one is carried straight down, so every prime ends up on the bottom row.




**1 Read the bottom row.**

a. Write the primes at the bottom, in order.

_____

b. Multiply them all together. What do you get? _____

## 2 Start it differently.

Build a new tree for 72, but split it as **8 × 9** at the top instead.

Did you get the same primes at the bottom? _____ In the same order? _____

## 3 Three more.

| Number | its primes multiplied out |
|---|---|
| 60 | |
| 100 | |
| 45 | |

**✎ ADD TO YOUR PROOF FILE**

Proof File page 5: both of your trees for 72 side by side, with the sentence **however you break it, the primes at the bottom are the same.**

**💬 TALK ABOUT IT**

Two people build different trees for the same number. Why can they never end up with different primes?

What is the factor tree of a prime number? Draw it.



**STUCK? TRY THIS**

It does not matter which factor pair you pick first. Split whatever is not yet prime, and keep going until every branch ends in a circled number.

**★ CHALLENGE ZONE**

A number has the prime factorization 2 × 2 × 2 × 3 × 5. What is the number?

Build the smallest number whose prime factorization uses exactly four primes, all different.

WEEK 6 NUMBER STRUCTURE

# Multiples That Meet

Two rhythms started together. When do they line up again?

**IN CLASS THIS WEEK**
Factors and multiples (4.OA.4)

**NUMBER WARM-UP · 2 MINUTES**
Count by 4s to 24. Now count by 6s to 24.

**WORDS TO KEEP**

**multiple** what you get by counting up in steps of a number

**common multiple** a number that is on both lists

**least common multiple** the smallest number on both lists

Two lists. Circle the first number that appears on both.

| count by 4 | 4 | 8 | 12 | 16 | 20 | 24 | 28 | 32 | 36 |
|---|---|---|---|---|---|---|---|---|---|
| count by 6 | 6 | 12 | 18 | 24 | 30 | 36 | 42 | 48 | 54 |

**1 Read it off.**

a. The first number on both lists: _____

b. The next two after that: _____ and _____

c. What do you notice about all the numbers on both lists?

_____

## 2 Do four more.

| Pair | least common multiple |
|---|---|
| 6 and 8 | |
| 5 and 7 | |
| 12 and 18 | |
| 4 and 12 | |

For one pair, the least common multiple is one of the two numbers. Which pair, and why?

## 3 Where the primes come in.

Write 12 and 18 as their primes. Now look at your answer of 36 as primes. Can you see where 36 came from?

**✎ ADD TO YOUR PROOF FILE**

Proof File page 6: your two lists with the meeting points circled, and what you noticed about every number that is on both.

**💬 TALK ABOUT IT**

Is there always a number on both lists? Say why you are sure.

When is the least common multiple just the two numbers multiplied together?



**STUCK? TRY THIS**

Write both lists out far enough to overshoot. The first shared number is the one you want, and every later one is a multiple of it.

**★ CHALLENGE ZONE**

Two buses leave the depot together at 9:00. One returns every 12 minutes, the other every 18. When are they next at the depot together?

Three buses now, every 4, 6 and 10 minutes, all leaving together at 9:00. When are all three next at the depot together?

WEEK 7 NUMBER STRUCTURE

# The Biggest Shared Factor

The other end of last week. Not where they meet, but what they share.

**IN CLASS THIS WEEK**
Factors and multiples (4.OA.4)

**NUMBER WARM-UP · 2 MINUTES**
Name every factor of 12. Now every factor of 18.

**WORDS TO KEEP**

**common factor** a number that divides both

**greatest common factor** the biggest number that divides both

Both factor lists, side by side. Circle every number that appears twice.

| factors of 24 | 1 | 2 | 3 | 4 | 6 | 8 | 12 | 24 | |
|---|---|---|---|---|---|---|---|---|---|
| factors of 36 | 1 | 2 | 3 | 4 | 6 | 9 | 12 | 18 | 36 |

**1 Read it off.**

a. Write every common factor.

_____

b. The greatest common factor is _____

## 2 Three more pairs.

| Pair | common factors | greatest |
|---|---|---|
| 18 and 30 | | |
| 15 and 28 | | |
| 48 and 60 | | |

One pair shares only the number 1. Which? _____

## 3 Use it for something.

You have **24** cookies and **36** brownies, and you want identical bags with nothing left over. What is the largest number of bags? _____ What goes in each bag?

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 7: the two factor lists with the shared ones circled, and the bag problem underneath, because that is what the greatest common factor is actually for.

**💬 TALK ABOUT IT**

Every pair of numbers shares at least one factor. Which one, and why always?

Last week you looked for where two lists meet. This week, where they overlap. Are those the same question?



**STUCK? TRY THIS**

List every factor of both numbers first, in full. The greatest common factor is the biggest number written on both lines, and it is often not the obvious one.

**★ CHALLENGE ZONE**

Two numbers whose greatest common factor is 1 are called **coprime**. Find three pairs of coprime numbers where neither number is prime.

A florist has 30 roses and 42 tulips and wants identical bunches. How many bunches, and what is in each?

WEEK 8 LOGIC & PROOF

# Coloring Proves It

Color the board first. The coloring does the arguing.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

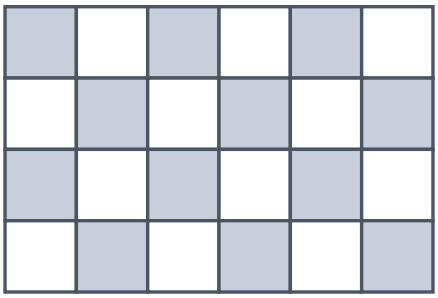
**NUMBER WARM-UP · 2 MINUTES**
A domino covers exactly two squares. On a colored board, how many of each color?

**WORDS TO KEEP**

**domino** a tile that covers exactly two squares side by side

**tiling** covering a board completely with no gaps and no overlaps

A colored board. Put one domino anywhere you like, going across or down.



**1 The one fact that matters.**

a. Your domino covers how many shaded squares? _____ How many white? _____

b. Try to place one that covers two shaded squares. Can you? _____ Why not?

_____

c. So 12 dominoes would cover _____ shaded and _____ white.

## 2 Now use it.

For each board, count the colors first. Then say whether dominoes could possibly cover it.

| Board | shaded | white | possible? |
|---|---|---|---|
| 3 by 3 | | | |
| 6 by 4 | | | |
| 5 by 5 | | | |
| 6 by 4 with two white squares cut out | | | |

## 3 Two different reasons.

Three of your boards are impossible. Two are impossible because of the **total** number of squares, and one because of the **colors**. Which is which?

✎ **ADD TO YOUR PROOF FILE**

Proof File page 8: a colored board with one domino drawn on it, and the sentence **every domino covers one of each color, always.** Everything next week rests on it.

💬 **TALK ABOUT IT**

The coloring is not part of the puzzle. Somebody added it. How can adding something prove anything?

If a board has equal colors, does that mean it definitely can be tiled?



**STUCK? TRY THIS**

Count the two colors before you try to place anything. If they are not equal, you can stop there, and if they are equal you still have to find an actual tiling.

★ **CHALLENGE ZONE**

A 5 by 5 board with the middle square removed leaves 24 squares. Count the colors. Does the coloring rule it out?

Then actually try to tile it. Compare what you found with what the coloring told you.

WEEK 9 LOGIC & PROOF

# The Domino Problem

Sixty-two squares, thirty-one dominoes, and it cannot be done.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

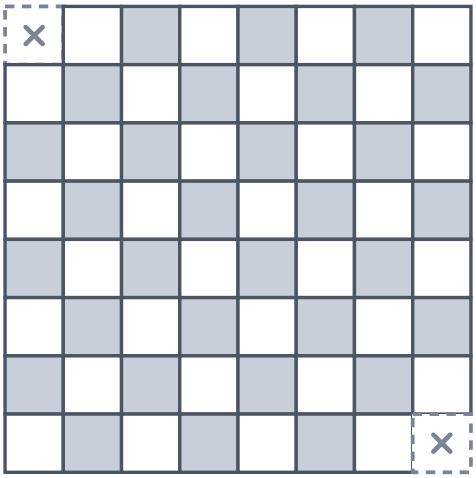
**NUMBER WARM-UP · 2 MINUTES**
How many squares on an 8 by 8 board? Take away 2.

**WORDS TO KEEP**

**counterexample** one case that shows a claim is false

**proof** an argument that covers every case at once, so no example is needed

An 8 by 8 board with two **opposite corners** cut out. Sixty-two squares are left, and you have thirty-one dominoes.



## 1 Finish the argument.

a. Both cut corners were the same color. Look at the squares beside them. Which color?

_____

b. So the board now has _____ shaded and _____ white.

c. Thirty-one dominoes would cover _____ of each color.

d. Write the contradiction in one sentence.

_____

## 2 A different pair of corners.

a. Now cut out the **top left** and the **top right** corners instead. What colors are they?

_____

b. Does the coloring rule this one out? _____ Is it actually possible? _____

## 3 State the rule.

Cut out any two squares you like. When is the board definitely impossible?

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 9. This is the best page in the book so far, so lay it out properly: the claim, the coloring, the two counts, and the contradiction. Four lines.

**💬 TALK ABOUT IT**

Nobody has ever tried all the ways to place thirty-one dominoes. Why do we not need to?

The second board is not ruled out by the coloring. Does that make it possible? What would you still have to do?



**STUCK? TRY THIS**

Count the shaded squares and the white squares separately, and then count what thirty-one dominoes must cover. The two numbers cannot both be right.

**★ CHALLENGE ZONE**

Cut out one shaded square and one white square, anywhere on the board, not just corners. Is it always still possible? Try several.

Now cut out four squares, two of each color, so that it becomes impossible anyway. The coloring is not the only thing that can go wrong.

WEEK 10 DOTS & LINES

# Degree Tells You

Add up every degree in any drawing at all. The answer is never odd.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

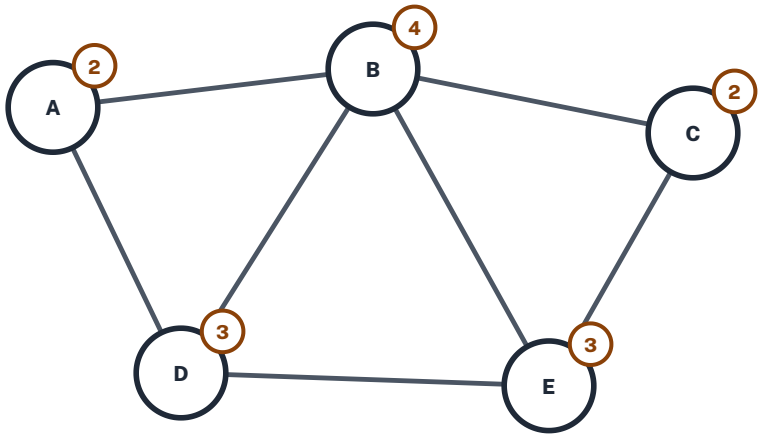
**NUMBER WARM-UP · 2 MINUTES**
A line has two ends. If you draw 5 lines, how many ends is that?

**WORDS TO KEEP**

**degree** how many lines meet at one dot

**theorem** a claim that has been proved, so it can be used without checking again

Every line has **two** ends, and each end lands on a dot. That single fact is the whole week.




**1 Count it twice.**

a. Add up all five degrees. _____

b. Count the lines. _____ Double it. _____

c. Write the theorem: the degrees always add up to _____

## 2 The second theorem, which is the useful one.

a. In the picture, how many dots have an **odd** degree? _____

b. Draw three graphs of your own and count the odd dots in each.

c. The number of odd dots is always _____

d. Why? The total is even. Every even degree adds an even amount. So the odd ones must come in _____

## 3 Now rule things out.

| Claim | impossible? | the reason |
|---|---|---|
| 7 houses, each joined to exactly 3 others | | |
| 6 houses, each joined to exactly 3 others | | |
| 9 people, each shook exactly 5 hands | | |

✎ **ADD TO YOUR PROOF FILE**

Proof File page 10: both theorems, each with the one-line reason. You will use the second one for the rest of the year and again in Grade 6.

💬 **TALK ABOUT IT**

Why is it impossible for exactly three dots to have odd degree?

Does the theorem tell you a graph exists when the count comes out even?



**STUCK? TRY THIS**

Do not draw the graph first. Multiply the number of dots by the degree each one is supposed to have. If that total is odd, stop, because no drawing can have an odd total.

★ **CHALLENGE ZONE**

Can you draw a graph with 5 dots where four of them have degree 3 and one has degree 2?

In any room of people, prove that the number who shook an odd number of hands is even. Grade 3 asked you to notice this. Now write the proof.

WEEK 11 DOTS & LINES

# Trace It or Prove You Cannot

The old bridge puzzle, settled in one line, without walking anywhere.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

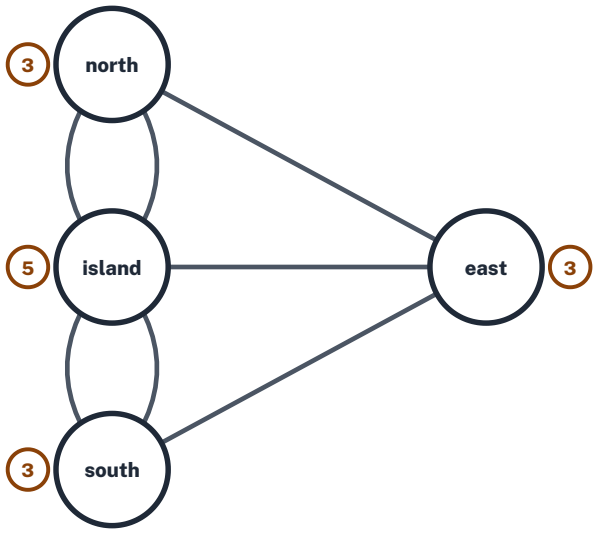
**NUMBER WARM-UP · 2 MINUTES**
How many odd numbers are in this list: 5, 3, 3, 3?

**WORDS TO KEEP**

**trace** draw the whole figure without lifting your pencil and without going over any line twice

**the tracing rule** a connected figure can be traced only if it has **0** or **2** odd dots. No other number works

Four pieces of land in the old city of Königsberg, joined by seven bridges. People tried for years to walk over every bridge exactly once.




## 1 Settle it in three lines.

a. Write the four degrees. _____

b. How many of them are odd? _____

c. The rule allows 0 or 2. So the walk is _____

## 2 Why the rule is what it is.

a. Think about a dot in the **middle** of your walk. You arrive on one line and leave on another, so its lines get used _____ at a time.

b. So a middle dot must have an _____ degree. Only the start and the finish can be odd, and that is why the answer is 0 or 2.

## 3 Four figures.

| Figure | odd dots | traceable? |
|---|---|---|
| a square | | |
| a square with one diagonal | | |
| a square with both diagonals | | |
| an envelope: a square, both diagonals and a roof | | |

**✎ ADD TO YOUR PROOF FILE**

Proof File page 11: the bridges, their four degrees, and the one sentence that ends the puzzle. Note that nobody has to walk anywhere.

**💬 TALK ABOUT IT**

People tried this walk for years. What did the rule give them that trying never could?

If a figure has 2 odd dots, where must you start?



**STUCK? TRY THIS**

Count the lines meeting at each dot and write the number beside it. Then count how many of those numbers are odd. That count is the entire answer.

**★ CHALLENGE ZONE**

Draw a figure with exactly 2 odd dots, then trace it. Where did you have to start and finish?

Add one more bridge to Königsberg so that the walk becomes possible. There is more than one answer, and each one changes two degrees at once.

WEEK 12 LOGIC & PROOF

# Impossible Fair

Six claims. Prove each one, or knock it down. You may not shrug.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

**NUMBER WARM-UP · 2 MINUTES**
No warm-up. Read all six first and start with whichever looks easiest.

**WORDS TO KEEP**

**prove** give a reason that covers every case

**disprove** give one single example where the claim fails. That is enough

Three of these are true and three are false. For a true one you must give the **reason**. For a false one, one example is enough to sink it.

## The claims

1. The sum of three odd numbers is always odd.
2. Every number with an odd number of factors is a square.
3. Seven people can each shake hands with exactly three others.
4. Every prime number is odd.
5. A 4 by 6 board with one square removed can be tiled with dominoes.
6. If a number is divisible by 2 and by 3, it is divisible by 6.

## 1 Your answers

| # | true or false | proof, or the example that kills it |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |

## 2 Which tool did each one need?

Write the tool beside each number: parity, factors, degree, coloring, primes.

_____

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 12: the three true claims with their reasons. Leave the false ones out of the file, but write the killing example beside each on a separate sheet.

**💬 TALK ABOUT IT**

Disproving takes one example and proving takes an argument. Why is that fair?

Claim 4 is knocked down by a single number. Does that make it a bad claim, or a useful one?



**STUCK? TRY THIS**

For each claim, ask first whether you believe it. If you do, hunt for the reason. If you do not, hunt for one example, and check it really does break the claim.

**★ CHALLENGE ZONE**

Write a seventh claim of your own that is false, but only just: it should take somebody a while to find the example. Swap with somebody else.

WEEK 13 LOGIC & PROOF

# Case Files

Everything from the first twelve weeks, in one folder.

**IN CLASS THIS WEEK**
Review and reasoning

**NUMBER WARM-UP · 2 MINUTES**
No warm-up this week.

Five cases. Each one needs the answer **and** the tool.

### 1 The locked cups

Nine cups upside down. You may turn over exactly two at a time. Can you get them all up? Which invariant decides it?

### 2 The shared factor

What is the greatest common factor of 42 and 70? And their least common multiple?

### 3 The village

Eleven houses, each to be joined to exactly five others. Possible?

### 4 The cut board

A 6 by 6 board with the top left and bottom right squares removed. Can 17 dominoes cover it?

_____

### 5 The tree

Write 84 as a product of primes. How many factors does 84 have altogether?

_____

### Which tool?

Beside each case number write its tool: parity invariant _____, common factors _____, degree _____, coloring _____, prime factorization _____.

**✎ ADD TO YOUR PROOF FILE**

Add a Proof File page called **My tools so far**: parity, factors and primes, degree, and coloring. One line each, in the words you would use out loud.

**💬 TALK ABOUT IT**

Three of these five cases say something cannot be done. Which three, and did you have to try a single arrangement to be sure?

Which tool has been the most useful so far? Which was hardest to trust?



**STUCK? TRY THIS**

Before reaching for a method, ask what is being **counted** and whether that count can change. Nearly every case this trimester turns on a count that cannot.

**★ CHALLENGE ZONE**

Invent a sixth case that looks like it needs coloring but actually needs parity. Write an answer key on a separate sheet.

WEEK 14 NUMBER STRUCTURE

# Divisible by 2, 5 and 10

Why does only the last digit matter? There is a reason, and it is short.

**IN CLASS THIS WEEK**
Place value to one million (4.NBT.1, 4.NBT.2)

**NUMBER WARM-UP · 2 MINUTES**
Is 3,470 divisible by 10? By 5? By 2?

**WORDS TO KEEP**

**divisible by** divides exactly, with remainder 0

**the reason** every place above the ones is worth a multiple of 10, and 10 is 2 × 5

Break **3,470** apart by place value. Look at what each piece is divisible by.

| Piece | value | divisible by 2? | by 5? | by 10? |
|---|---|---|---|---|
| 3 thousands | 3000 | | | |
| 4 hundreds | 400 | | | |
| 7 tens | 70 | | | |
| 0 ones | 0 | | | |

**1 So the rule falls out.**

a. Every piece except the **ones** is divisible by 10, and therefore by _____ and by _____.

b. So the whole number is divisible by 2 exactly when the _____ is.

## 2 Test them.

| Number | by 2 | by 5 | by 10 | by 4 |
|---|---|---|---|---|
| 3,470 | | | | |
| 8,216 | | | | |
| 9,005 | | | | |
| 12,348 | | | | |

## 3 Why 4 needs two digits.

a. Is 100 divisible by 4? _____ Is 1,000? _____

b. So for 4, the **hundreds** and everything above them take care of themselves, and you only need the last _____ digits.

✎ **ADD TO YOUR PROOF FILE**

Proof File page 13: the three rules for 2, 5 and 10, each with its reason in one line. A rule without its reason does not go in the file.

💬 **TALK ABOUT IT**

Why is there no last-digit rule for 3? Test 12 and 21 and 13 to see the trouble.

What would the rule for 8 be, and how many digits would you need?



**STUCK? TRY THIS**

Split the number into thousands, hundreds, tens and ones, and check each piece separately. Every piece that is already divisible can be ignored.

★ **CHALLENGE ZONE**

Work out the rule for 8 and test it on 1,048 and on 3,036.

A number is divisible by 20. What must be true about its last two digits?

WEEK 15 NUMBER STRUCTURE

# Divisible by 3 and 9

Add the digits. It is not a trick, and here is why it works.

**IN CLASS THIS WEEK**
Place value and multi-digit arithmetic (4.NBT.1, 4.NBT.4)

**NUMBER WARM-UP · 2 MINUTES**
Add the digits of 4,275. Is that total divisible by 9?

**WORDS TO KEEP**

**digit sum** add all the digits of a number

**the reason** 10 is 9 + 1, and 100 is 99 + 1, and 1,000 is 999 + 1. The 9s and 99s always divide by 9, so only the leftover 1s matter

Take **4,275** apart, and split each place into a run of nines plus a leftover.

| Place | written out | the nines part | the leftover |
|---|---|---|---|
| 4 thousands | 4 × 999 + 4 | 4 × 999 | 4 |
| 2 hundreds | 2 × 99 + 2 | 2 × 99 | 2 |
| 7 tens | 7 × 9 + 7 | 7 × 9 | 7 |
| 5 ones | 5 | 0 | 5 |

**1 Read the last column.**

a. The leftovers are 4, 2, 7, 5. What are those? _____

b. Every nines part divides by 9. So 4,275 is divisible by 9 exactly when _____ is.

## 2 Test four numbers.

| Number | digit sum | by 3? | by 9? |
|---|---|---|---|
| 4,275 | | | |
| 1,236 | | | |
| 8,001 | | | |
| 5,555 | | | |

A number divisible by 9 is always divisible by 3. Is the reverse true? _____

## 3 Put two rules together.

A number is divisible by 6 exactly when it is divisible by _____ and by _____. Test 1,236 and 8,001 for divisibility by 6.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 14: the digit-sum rule with the 9 + 1 reason. This is the page next week depends on, so write the reason out in full.

**💬 TALK ABOUT IT**

Why does the same digit sum work for both 3 and 9?

If you scramble the digits of a number, does it stay divisible by 9? Say why.



**STUCK? TRY THIS**

You may add the digits again if the total is still big. 4 + 2 + 7 + 5 = 18, and 1 + 8 = 9, which is easier to judge.

**★ CHALLENGE ZONE**

Is 123,456,789 divisible by 9? Do it in your head.

Find the smallest number greater than 1,000 that is divisible by both 8 and 9.

WEEK 16 NUMBER STRUCTURE

# Casting Out Nines

A real check on multiplication, from the rule you proved last week.

**IN CLASS THIS WEEK**
Multi-digit multiplication (4.NBT.5)

**NUMBER WARM-UP · 2 MINUTES**
Digit sum of 27? Of 34? Keep adding until one digit is left.

**WORDS TO KEEP**

**casting out nines** replacing a number by its digit sum, over and over, until one digit is left

**a check** a test that catches wrong answers. It does not promise a right one

Somebody worked out **27 × 34 = 918**. Check it without multiplying again.

| | the number | cast down to one digit |
|---|---|---|
| first factor | 27 | |
| second factor | 34 | |
| multiply those two | | |
| their answer | 918 | |

**1 Compare the last two rows.**

a. Do they match? _____

b. If they had **not** matched, what would you know for certain?

_____

## 2 Three answers. One is wrong.

| Multiplication | left side cast | answer cast | wrong? |
|---|---|---|---|
| 43 × 26 = 1,118 | | | |
| 58 × 37 = 2,156 | | | |
| 64 × 45 = 2,880 | | | |

Which one is definitely wrong? _____ Work out the right answer. _____

## 3 What the check cannot do.

Somebody wrote 918 as 981 by mistake. Cast both down. _____ and _____. Does the check spot the error? _____ Say why not.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 15: the check written as three steps, and underneath it, honestly, the kind of mistake it will never catch.

**💬 TALK ABOUT IT**

A check that passes does not mean the answer is right. Is a check like that worth having?

Roughly how many wrong answers would slip past it? Think about how many possible single digits there are.



**STUCK? TRY THIS**

Cast each factor down to one digit, multiply those two small numbers, then cast that down too. Compare it with the answer cast down. Two different numbers means a mistake.

**★ CHALLENGE ZONE**

Make up a multiplication, get it deliberately wrong, and check whether casting out nines catches you.

Does the same check work for addition? Try it on three sums and say what you find.

WEEK 17 NUMBER STRUCTURE

# Wrapping Around, Again

Casting out nines was not a trick. It was arithmetic on a wheel.

**IN CLASS THIS WEEK**
Multi-digit arithmetic and patterns (4.NBT.4, 4.OA.5)

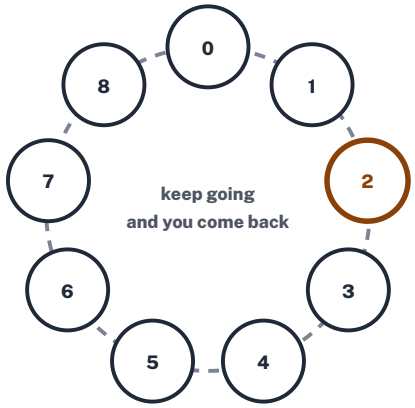
**NUMBER WARM-UP · 2 MINUTES**
What is the remainder when 20 is divided by 9? When 38 is?

**WORDS TO KEEP**

**mod** short for the remainder after dividing. 38 mod 9 is 2

**the wheel** the numbers 0 to 8 in a ring. Counting past 8 lands back on 0

The nine remainders, in a ring. Every number in the world sits on one of these nine places.




## 1 Put numbers on the wheel.

| Number | ÷ 9 leaves | Number | ÷ 5 leaves |
|---|---|---|---|
| 38 | | 38 | |
| 100 | | 100 | |
| 918 | | 918 | |

Compare your 918 answer with its digit sum from last week. The wheel has no 9: a digit sum of 9 sits on the 0. _____

## 2 Why the two weeks are the same week.

Casting out nines replaces a number by its digit sum. The wheel replaces it by its remainder after 9. Write one sentence saying why those give the same place on the wheel.

## 3 Powers on a wheel.

Work out the **last digit** of each, which is the same as working mod 10.

| | 3 | 3 × 3 | × 3 | × 3 | × 3 | × 3 |
|---|---|---|---|---|---|---|
| last digit | 3 | 9 | | | | |

How long before the pattern repeats? _____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 16: the wheel of nine, with 918 marked on it and the digit-sum connection written underneath.

**💬 TALK ABOUT IT**

Why does every number land somewhere on the wheel, however big it is?

The last digit of 3, 9, 27, 81 repeats. What is the last digit of twenty 3s multiplied together? You do not need to multiply it all out.



**STUCK? TRY THIS**

Divide and keep only the remainder. Everything else about the number can be thrown away, which is exactly what makes the wheel useful.

**★ CHALLENGE ZONE**

What is the last digit of thirty 7s multiplied together? Find the cycle first.

Which numbers land on 0 on the nine-wheel? Describe them without listing them.

WEEK 18 NUMBER STRUCTURE

# The Day of the Week

A whole year on a seven-wheel, and no counting on fingers.

**IN CLASS THIS WEEK**
Multi-step problems; measurement (4.OA.3, 4.MD.2)

**NUMBER WARM-UP · 2 MINUTES**
What is 59 ÷ 7? What is the remainder?

**WORDS TO KEEP**

**offset** how many days a month starts after January 1st

**the seven-wheel** only the remainder after dividing by 7 changes the day

In a year that is **not** a leap year, **January 1st is a Monday**. The offset is how many days later each month begins.

| Month | offset | ÷ 7 leaves | starts on |
|---|---|---|---|
| January | 0 | | |
| February | 31 | | |
| March | 59 | | |
| April | 90 | | |
| May | 120 | | |
| June | 151 | | |

**1 Do March first.**

a. March 1st is 59 days after January 1st. 59 ÷ 7 = _____ remainder _____

b. Move that many days on from Monday. _____

## 2 Finish the second half of the year.

| Month | offset | ÷ 7 leaves | starts on |
|---|---|---|---|
| July | 181 | | |
| August | 212 | | |
| September | 243 | | |
| October | 273 | | |
| November | 304 | | |
| December | 334 | | |

## 3 Look down the last column.

a. Which month starts on the same day as **January**? _____

b. Find two more months that share a starting day with each other.

_____

✎ **ADD TO YOUR PROOF FILE**

Proof File page 17: the twelve offsets with their remainders after 7. Any year that starts on a Monday and is not a leap year works exactly the same way.

💬 **TALK ABOUT IT**

The offsets grow to 334 and the answers only ever take 7 values. Why?

In a leap year, which months would shift and which would not?



**STUCK? TRY THIS**

Divide the offset by 7 and throw away the whole weeks. Only the remainder moves you forward from Monday.

★ **CHALLENGE ZONE**

February 1st and March 1st start on the same day in a normal year. Why does that happen, and what does it tell you about the length of February?

In a leap year, does that pair still match? Work it out rather than guessing.

WEEK 19 SHAPE & SPACE

# Lines of Symmetry

Fold it. If the two halves land on each other, you found a line.

**IN CLASS THIS WEEK**
Lines of symmetry in two-dimensional figures (4.G.3)

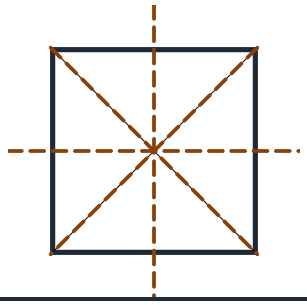
**NUMBER WARM-UP · 2 MINUTES**
How many ways can you fold a square in half so the halves match?

**WORDS TO KEEP**

**line of symmetry** a fold line where one half lands exactly on the other

**regular** a shape with all sides equal and all corners equal

A square has four fold lines. Two go through the middles of the sides and two go corner to corner.



**1 Check each one.**

a. Fold along the up-and-down line. Does every corner land on a corner? _____

b. Fold corner to corner. Same question. _____

c. Try a fifth line, anywhere else. What goes wrong?

_____

## 2 Count the lines for each shape.

| Shape | lines of symmetry |
|---|---|
| square | 4 |
| rectangle that is not a square | |
| equilateral triangle | |
| isosceles triangle that is not equilateral | |
| regular hexagon | |
| regular pentagon | |
| parallelogram that is not a rectangle | |
| scalene triangle | |

## 3 Two of them got zero.

Both zeroes surprise people, because the shapes still look balanced. Draw the parallelogram and show a fold that **nearly** works, then say exactly what goes wrong.

✎ **ADD TO YOUR PROOF FILE**

Proof File page 18: the eight shapes with their line counts. Circle the two that have none, because next week is about them.

💬 **TALK ABOUT IT**

A parallelogram looks symmetrical and has no fold line. What is your eye seeing?

Does a shape with more sides always have more lines of symmetry?



**STUCK? TRY THIS**

Trace the shape, cut it out and actually fold it. A line that only nearly works does not count, and folding is the fastest way to find that out.

★ **CHALLENGE ZONE**

A regular polygon with 7 sides. How many lines of symmetry? Now one with 12 sides. Write the rule.

Draw a shape with exactly two lines of symmetry that is **not** a rectangle.

WEEK 20 SHAPE & SPACE

# Turning a Shape

Fold lines are not the only kind of symmetry. Some shapes only turn.

**IN CLASS THIS WEEK**
Lines of symmetry; angle measure (4.G.3, 4.MD.5)

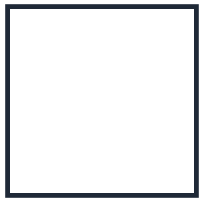
**NUMBER WARM-UP · 2 MINUTES**
A quarter turn is how many degrees? A half turn?

**WORDS TO KEEP**

**turn symmetry** turning a shape less than all the way around and having it land on itself

**order** how many turns, including the full turn back to the start, land it on itself

Turn a square a quarter turn. It lands on itself. Keep going and count how many landings you get before you are back where you started.



## 1 Turn it around.

| Turn | 0 | quarter | half | three quarters | full |
|---|---|---|---|---|---|
| lands on itself? | yes | | | | |

So the square has turn symmetry of order _____

## 2 Fill both columns.

| Shape | fold lines | turn order |
|---|---|---|
| square | 4 | 4 |
| rectangle, not a square | | |
| equilateral triangle | | |
| regular hexagon | | |
| parallelogram, not a rectangle | | |
| isosceles triangle, not equilateral | | |

## 3 The interesting row.

a. Which shape has turn symmetry but **no** fold lines at all? _____

b. So the two kinds of symmetry are not the same thing. Is there a shape with fold lines but no turn symmetry beyond the full turn? _____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 19: the two-column table. The parallelogram row is the point of the page, so mark it.

**💬 TALK ABOUT IT**

Why does every shape have turn order at least 1?

A shape has 5 fold lines. What do you expect its turn order to be? Check with the pentagon.



**STUCK? TRY THIS**

Trace the shape on tracing paper, put a pin through the middle and turn the tracing. Count how many times the tracing fits the drawing underneath.

**★ CHALLENGE ZONE**

Draw a shape with turn order 3 and no fold lines. A pinwheel is a good starting point.

For a regular polygon, how are the number of fold lines and the turn order related? Say why.

WEEK 21 SHAPE & SPACE

# The Eight Ways to Move a Square

Every move that leaves a square looking the same. There are exactly eight.

**IN CLASS THIS WEEK**
Angle measure and reasoning (4.MD.5, 4.OA.3)

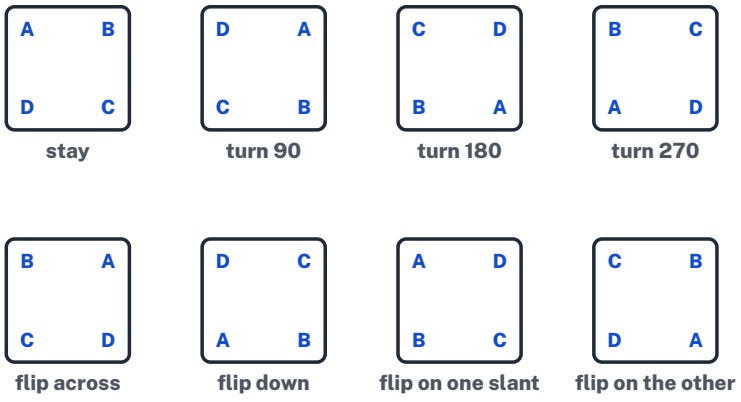
**NUMBER WARM-UP · 2 MINUTES**
A quarter turn and then another quarter turn is the same as what?

**WORDS TO KEEP**

**a move** here it means turning or flipping a square so it lands exactly on itself

**composing** doing one move and then another. The result is always one of the eight

The corners are labeled A, B, C, D so you can see where each one goes. Four turns and four flips, and that is all of them.




## 1 Check the count.

a. How many turns? _____ How many flips? _____ Total: _____

b. Is there a ninth? Try to invent one, then find it in the picture.

_____

## 2 Do two in a row.

Follow the corner letters through both moves, then find the single move that would have done the same thing.

| first move | then this move | same as |
|---|---|---|
| turn 90 | turn 90 | |
| turn 90 | turn 270 | |
| flip across | flip down | |
| flip across | flip across | |

## 3 Undo.

a. Which moves, other than **stay**, get you back to the start when you do them **twice**? List them.

_____

b. Which move undoes **turn 90**? _____

✎ **ADD TO YOUR PROOF FILE**

Proof File page 20: the eight moves listed, and the fact that doing any two of them in a row always lands you on another one of the eight. That closed-ness is the surprise.

💬 **TALK ABOUT IT**

Doing two moves always gives a move already on the list. Why could it not give a ninth?

Which single move is the odd one out, in the sense that it changes nothing?



**STUCK? TRY THIS**

Track one corner, say A, all the way through. Where A ends up plus whether the square got flipped over is enough to name the move.

★ **CHALLENGE ZONE**

How many moves does an **equilateral triangle** have? Count turns and flips.

And a rectangle that is not a square? The answer is smaller than eight, and working out which moves are lost is the puzzle.

WEEK 22 SHAPE & SPACE

# Angles That Must Add Up

Tear the three corners off any triangle at all. They make a straight line.

**IN CLASS THIS WEEK**
Angle measure and adding angles (4.MD.5, 4.MD.7)

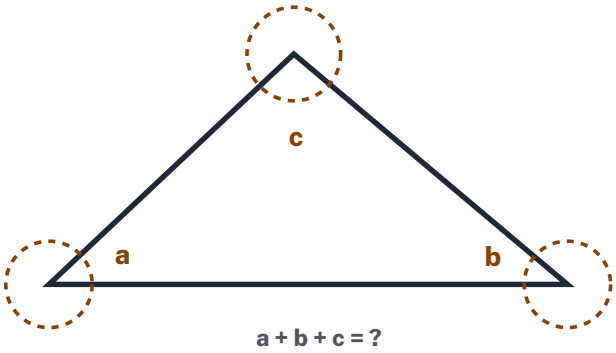
**NUMBER WARM-UP · 2 MINUTES**
How many degrees in a straight line? In a quarter turn?

**WORDS TO KEEP**

**degree** the unit for measuring turn. A full turn is 360 degrees

**straight angle** half a turn, which is 180 degrees

Draw any triangle you like on scrap paper. Tear off all three corners and put them together, points touching.




**1 What you get.**

a. The three corners together make a _____

b. So a + b + c = _____ degrees.

c. Do it again with a completely different triangle. Same answer? _____

## 2 Find the missing angle.

| a | b | c | possible? |
|---|---|---|---|
| 60 | 70 | | |
| 90 | 45 | | |
| 30 | 30 | | |
| 90 | 95 | | |

One row is impossible. Which, and why?

_____

## 3 More than three sides.

a. Draw a four-sided shape and cut it into **two** triangles with one straight line. So its angles add to _____

b. A five-sided shape splits into _____ triangles, so its angles add to _____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 21: the torn-corner picture, the 180, and the splitting-into-triangles trick that turns it into a rule for any shape.

**💬 TALK ABOUT IT**

The tearing works for every triangle anybody has ever drawn. Is that a proof?

Can a triangle have two right angles? Use the rule rather than trying to draw one.



**STUCK? TRY THIS**

For a shape with more sides, draw all the lines from one corner. Count the triangles you made, then multiply by 180.

**★ CHALLENGE ZONE**

Work out the angle total for a six-sided shape and a ten-sided shape.

Write the rule for a shape with **n** sides. You will need to say how many triangles it splits into first.

WEEK 23 SHAPE & SPACE

# Tiling the Floor

Only three regular shapes tile a floor. The reason is one division.

**IN CLASS THIS WEEK**
Angle measure (4.MD.5, 4.MD.7)

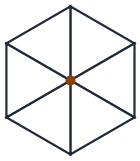
**NUMBER WARM-UP · 2 MINUTES**
What is 360 ÷ 60? What is 360 ÷ 90? What is 360 ÷ 108?

**WORDS TO KEEP**

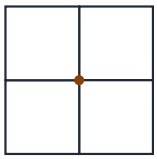
**tile** cover a flat floor with copies of one shape, no gaps and no overlaps

**corner angle** the angle inside a shape at one corner. Copies meeting at a point must have corner angles adding to exactly 360

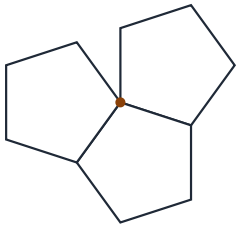
Put copies of one shape around a single point. They fit only if the corner angles add to exactly **360**.



each corner is 60 degrees
**6 of them fit, they close up exactly**



each corner is 90 degrees
**4 of them fit, they close up exactly**



each corner is 108 degrees
**3 of them fit, 36 degrees left over**

**1 Read the three pictures.**

Triangles fit _____ at a point, squares fit _____, and pentagons leave _____ degrees over.

## 2 Do the division for each shape.

| Regular shape | corner angle | 360 ÷ that | tiles? |
|---|---|---|---|
| triangle | 60 | | |
| square | 90 | | |
| pentagon | 108 | | |
| hexagon | 120 | | |
| octagon | 135 | | |

## 3 Why it stops.

a. Which three shapes tile? _____

b. A shape with more than six sides has a corner angle bigger than 120 and smaller than 180. So 360 divided by it lands between _____ and _____, and can never be a whole number. That is the whole proof.

**✎ ADD TO YOUR PROOF FILE**

Proof File page 22: the three shapes that tile, the division that decides it, and the argument that nothing with seven or more sides can ever work.

**💬 TALK ABOUT IT**

Bathroom floors are usually square tiles and honeycombs are hexagons. Does the rule explain both?

Pentagons leave 36 degrees over. Could you fill that gap with a different shape?



**STUCK? TRY THIS**

Work out the corner angle first, then divide 360 by it. Only a whole-number answer lets the copies close up.

**★ CHALLENGE ZONE**

Two regular octagons meeting at a point use 135 + 135 degrees. What angle is left, and which regular shape fits it exactly? Real bathroom floors are tiled this way.

Any triangle at all, even a lopsided one, tiles a floor. Try it with six copies of a scalene triangle and say why it works when regular pentagons do not.

WEEK 24 SHAPE & SPACE

# Impossible Tilings

Two boards, one corner different, and one coloring settles only one of them.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

**NUMBER WARM-UP · 2 MINUTES**
Is 64 divisible by 3? Is 63?

**WORDS TO KEEP**

**counting argument** the number of squares must divide by the size of the tile, or stop right there

**coloring argument** when counting is not enough, color the board so that every tile must cover one of each color

The first question is always the easy one: does the number of squares even divide by the size of the tile?

| Board | squares | tile | divides? | possible? |
|---|---|---|---|---|
| 5 by 5 | 25 | domino (2) | | |
| 4 by 4 | 16 | L-tromino (3) | | |
| 8 by 8 | 64 | 1 by 3 (3) | | |
| 6 by 4 | 24 | 1 by 3 (3) | | |

**1 When counting is not enough.**

The last row divides exactly. Does that mean it can be done? _____ What would you still have to do?

_____

## 2 Color it in threes.

Number every square by **row + column**, then color it by the remainder after 3. A 1 by 3 tile, lying either way, always covers one of each color.

a. On a full 8 by 8 board the three color counts are **21**, **22** and **21**. Are they equal? _____

b. That already rules out the whole board, but so did counting. Now take one square away, leaving 63.

## 3 The two corners.

a. Remove the **top left** corner. It is a color there are 21 of. The counts become 20, 22, 21. Equal? _____ So it is _____

b. Remove the **top right** corner instead. It is the color there are 22 of, so the counts become 21, 21, 21. Does the coloring rule this one out? _____

c. Now flip the whole board over, left to right. Where does the missing top right corner land? So is the second board possible? _____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 23: the two corners side by side. Same board, same tiles: the coloring settles the first, and flipping the board settles the second. Write down both reasons.

**💬 TALK ABOUT IT**

The coloring rules out one corner and says nothing about the other. Does saying nothing mean the second one is possible?

Why does a 1 by 3 tile always cover one of each color, whichever way it lies?



**STUCK? TRY THIS**

Do the counting test first, because it is quick and it settles most boards. Only reach for a coloring when the count divides exactly and you still cannot do it.

**★ CHALLENGE ZONE**

Color a 5 by 5 board like Week 8, with the corners shaded. Take away one square so that dominoes can cover the rest. Which squares work?

A 9 by 9 board has 81 squares and its three color counts are all 27. What does that tell you, and what does it not tell you?

WEEK 25 LOGIC & PROOF

# Puzzle Fair

A little of everything from Weeks 14 to 24.

**IN CLASS THIS WEEK**
Review and reasoning

**NUMBER WARM-UP · 2 MINUTES**
No warm-up. Start wherever you like.

Six puzzles. Answer plus tool, every time.

### 1 The long division

Is 71,325 divisible by 3? By 9? By 5? Do all three without dividing.

### 2 The bad answer

Somebody says 46 × 27 = 1,232. Check it by casting out nines.

### 3 The folded shape

A regular shape has 6 fold lines. What is its turn order, and what shape is it?

### 4 The floor

Could a floor be tiled with regular nine-sided shapes? Find the corner angle first.

### 5 The calendar

In a year starting on Monday that is not a leap year, what day is December 1st? Its offset is 334.

### 6 The cut board

A 7 by 7 board has one corner removed. Can dominoes cover the remaining 48 squares? Count the colors, and if they do not rule it out, find a covering.

**✎ ADD TO YOUR PROOF FILE**

Add a Proof File page called **My tools, weeks 14 to 24**: divisibility with reasons, casting out nines, the seven-wheel, symmetry, corner angles, and coloring.

**💬 TALK ABOUT IT**

In puzzle 6 the count and the coloring both come out even. Does that settle it? What does?

Which puzzle could you do entirely in your head? What made it easy?



**STUCK? TRY THIS**

Almost everything this trimester is either a remainder or a count of two colors. Work out which one a puzzle is asking about before you start.

**★ CHALLENGE ZONE**

Write a seventh puzzle that looks like a divisibility question but is really about symmetry. Include an answer key on a separate sheet.

# Choosing a Team

Line them up, then divide out the orderings you did not want.

**IN CLASS THIS WEEK**
Multi-step problems (4.OA.3)

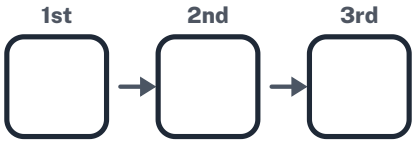
**NUMBER WARM-UP · 2 MINUTES**
What is 5 × 4 × 3? What is 3 × 2 × 1?

**WORDS TO KEEP**

**arrangement** an order. Ana then Ben is different from Ben then Ana

**combination** a choice where the order does not matter, so those two are one team

Five children, and **three** of them go on the trip. Count it in two steps: first count the line-ups, then divide out the ones that are the same team.




first pick, then second, then third

**1 Step one, then step two.**

a. How many for the first pick? _____ Then _____, then _____. Multiply: _____

b. Take one team, say Ana, Ben, Cal. How many different orders could those same three be picked in? _____

c. So divide. Number of teams: _____

## 2 Four more.

| Choose | line-ups | orders of one team | teams |
|---|---|---|---|
| 2 from 6 | 6 × 5 = 30 | 2 | |
| 3 from 6 | | | |
| 4 from 6 | | | |
| 3 from 7 | | | |

## 3 Two of your answers match.

Choosing 2 from 6 and choosing 4 from 6 give the same number. That is not a coincidence. Say why, thinking about the children you **leave behind**.

✎ **ADD TO YOUR PROOF FILE**

Proof File page 24: the two-step method, and the leaving-behind reason for why choosing 2 and choosing 4 from the same six come out equal.

💬 **TALK ABOUT IT**

Why do you divide rather than subtract?

Choosing 0 children from 6: how many ways? The answer is not 0.



**STUCK? TRY THIS**

Count the line-ups first and do not worry that you are over-counting. Then work out exactly how many times each team got counted, and divide by that.

★ **CHALLENGE ZONE**

How many ways to choose 3 from 8? And 5 from 8? Check that the leaving-behind rule predicts the second from the first.

A pizza shop has 7 toppings and you pick 3. How many pizzas?

# The Triangle Grows

Every number is the two above it added together, and that rule builds the whole thing.

**IN CLASS THIS WEEK**
Number and shape patterns (4.OA.5)

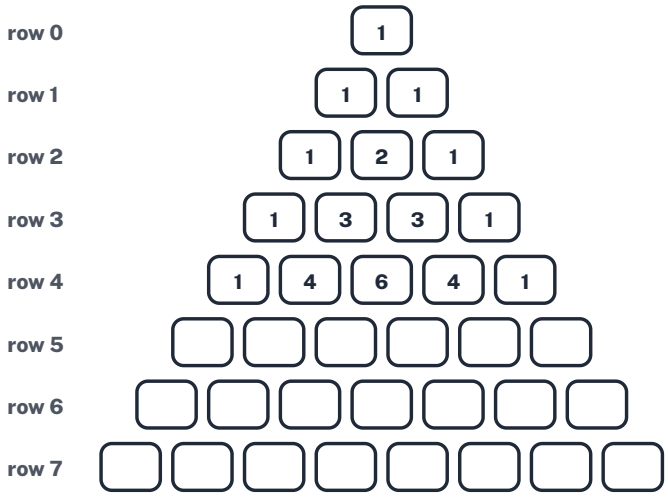
**NUMBER WARM-UP · 2 MINUTES**
1 + 4 = ?  4 + 6 = ?  10 + 10 = ?

**WORDS TO KEEP**

**recurrence** a rule that builds the next line out of the line before it

**row** the triangle's rows are numbered from 0, so row 4 is the fifth line down

You met this triangle in Grade 3. This year you build it from its rule and then use it. Rows 0 to 4 are filled in.




**1 Build three more rows.**

a. Row 5: 1, _____, _____, _____, _____, 1

b. Fill in rows 6 and 7 on the picture.

## 2 Two patterns to find.

a. Add up each row. Write the totals for rows 0 to 7.

_____

b. Each total is the one before it _____

c. Read down the third number in each row: 1, 3, 6, 10, 15, 21. Where did you meet those in Grade 3? _____

## 3 Why the edges are always 1.

A number on the edge has only **one** number above it, not two. Explain what the missing one counts as.

_____

✎ **ADD TO YOUR PROOF FILE**

Proof File page 25: rows 0 to 7 of the triangle with the row totals down one side. Next week connects this page to last week's teams.

💬 **TALK ABOUT IT**

The rule only ever adds. So why do the numbers grow so fast?

Row 7 adds up to 128. Predict the total of row 10 without building it.



**STUCK? TRY THIS**

Work one row at a time, left to right, and write each number under the gap between the two you added. The edges are always 1.

★ **CHALLENGE ZONE**

Build rows 8, 9 and 10.

The biggest number in row 10 is in the middle. What is it, and how does it compare with the row total?

# The Triangle and the Teams

Last week's triangle and the week before's teams are the same numbers.

**IN CLASS THIS WEEK**
Number and shape patterns (4.OA.5)

**NUMBER WARM-UP · 2 MINUTES**
How many ways to choose 2 children from 4? Look at row 4 of the triangle.

**WORDS TO KEEP**

**row n** the row of the triangle that starts 1, then n

**choosing k from n** the number of teams of k you can pick out of n people

Row 6 of the triangle is **1, 6, 15, 20, 15, 6, 1**. Underneath it, work out the teams you can choose from six children.

| Choose from 6 | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| how many teams | | | | | | | |
| row 6 says | 1 | 6 | 15 | 20 | 15 | 6 | 1 |

**1 They match.**

a. Choosing 2 from 6 gave you 15 last week. Where is 15 in row 6? _____

b. Choosing **0** from 6 is 1 way, which is doing nothing. Does that fit the row? _____

## 2 Read answers straight off the triangle.

| Question | row | position | answer |
|---|---|---|---|
| choose 3 from 6 | 6 | 3 | |
| choose 2 from 7 | | | |
| choose 3 from 7 | | | |
| choose 4 from 8 | | | |

## 3 What the row total counts.

a. Row 6 adds up to _____

b. Each child either comes or stays home, so there are 2 choices each, six times over: 2 × 2 × 2 × 2 × 2 × 2 = _____. Same number. Say why.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 26: row 6 with the team counts written underneath it, and the come-or-stay-home argument for the row total.

**💬 TALK ABOUT IT**

Why does the triangle answer a question about teams when it was built by adding?

Row 6 reads the same forwards and backwards. Which fact about teams is that?



**STUCK? TRY THIS**

The positions in a row are counted from 0, not from 1. Choosing 2 from 6 is the number in **position 2** of **row 6**.

**★ CHALLENGE ZONE**

Use row 8 to answer: how many ways to choose 3 from 8, and how many to choose 5 from 8?

A row of the triangle adds up to 1,024. Which row is it?

# Trees

Joined up, with no way around. Count the lines and you will find a rule.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

**NUMBER WARM-UP · 2 MINUTES**
5 dots joined up with no loops. Guess how many lines before you count.

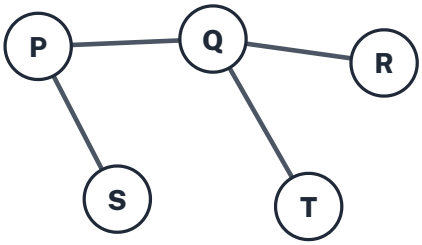
**WORDS TO KEEP**

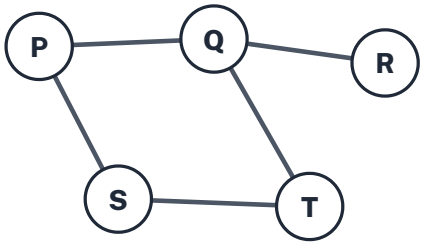
**connected** you can get from any dot to any other along the lines

**cycle** a path that leaves a dot and comes back to it without repeating a line

**tree** connected, and with no cycle anywhere in it

The first drawing is a tree. The second is not, and one extra line is the whole difference.




**1 Spot the difference.**

a. Find the cycle in the second drawing and write the dots it goes through.

_____

b. Dots and lines in the tree: _____ and _____

c. Dots and lines in the other one: _____ and _____

## 2 Draw your own and count.

| Dots in the tree | lines |
|---|---|
| 3 | |
| 5 | |
| 8 | |
| 12 | |

Write the rule: a tree with **n** dots has _____ lines.

## 3 Why one fewer.

Start with one dot and no lines. Every new dot you add has to be joined on by exactly **one** line, or it would either be cut off or make a cycle. Finish the argument.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 27: the rule, and the one-dot-at-a-time argument for why it is always true. It is another invariant, in disguise.

**💬 TALK ABOUT IT**

Why would a second line to a new dot make a cycle?

Is every graph with 5 dots and 4 lines a tree? Draw one that is not.



**STUCK? TRY THIS**

Add the dots one at a time and watch the count. Each new dot brings exactly one new line with it, which is why the lines always stay one behind.

**★ CHALLENGE ZONE**

A graph has 10 dots and 8 lines. Can it be connected? Use the rule.

A graph has 10 dots and 12 lines and is connected. How many cycles must it have at least?

# The Cheapest Network

Join every town for the least money. The answer is always a tree.

**IN CLASS THIS WEEK**
Multi-step problems (4.OA.3)

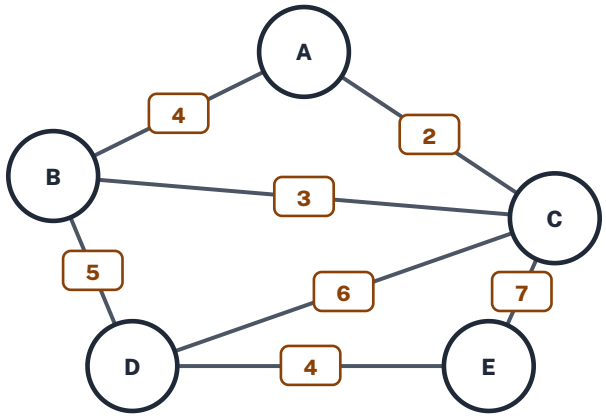
**NUMBER WARM-UP · 2 MINUTES**
Which is cheaper, 2 + 3 + 4 + 5 or 2 + 3 + 4 + 6?

**WORDS TO KEEP**

**network** the set of roads you decide to build

**cheapest network** joins every town for the smallest total cost

Five towns. The number on each road is what it costs to build. You must be able to get from every town to every other, for as little as possible.




## 1 Build it cheapest first.

The roads are already listed cheapest first. Take each one **unless** it would make a cycle.

| Road | cost | take it? | why not |
|---|---|---|---|
| A to C | 2 | | |
| B to C | 3 | | |
| A to B | 4 | | |
| D to E | 4 | | |
| B to D | 5 | | |
| C to D | 6 | | |
| C to E | 7 | | |

## 2 Check your answer.

a. How many roads did you build? _____ How many towns are there? _____

b. Is that the tree rule from last week? _____

c. Total cost: _____

## 3 Why a cycle is always a waste.

Suppose your network had a cycle in it. You could delete the most expensive road on that cycle and still get everywhere. Say why, and say what that proves about the cheapest network.

_____

**✎ ADD TO YOUR PROOF FILE**

Proof File page 28: your network drawn in, its total cost, and the sentence **the cheapest network can never contain a cycle.**

**💬 TALK ABOUT IT**

You never looked at all the possible networks. Why is taking the cheapest road each time good enough?

If two roads cost the same, does it matter which you take first?



**STUCK? TRY THIS**

Sort the roads by cost before you start. Then take them in order, skipping any road whose two towns are already joined by roads you have taken.

**★ CHALLENGE ZONE**

Change the cost of B to D from 5 to 1 and rebuild. Which roads change?

If every road cost exactly the same, how many different cheapest networks would there be?

WEEK 31 STEPS & RULES

# Two Ways to Multiply

Two methods, the same answer, and a fair way to compare the work.

**IN CLASS THIS WEEK**
Multiplying multi-digit numbers (4.NBT.5)

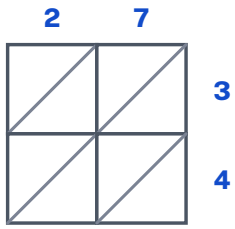
**NUMBER WARM-UP · 2 MINUTES**
What is 7 × 4? 2 × 3? Those are single-digit multiplications.

**WORDS TO KEEP**

**single-digit multiplication** one times-table fact, like 7 × 4. The unit of work for this week

**lattice** a grid method where every times-table fact gets its own box

The lattice for **27 × 34**. Each box takes one digit from the top and one from the side, and the answer goes in split by the diagonal.




27 × 34

**1 Fill it in.**

a. How many boxes are there? _____ So how many times-table facts do you need?

_____

b. Fill in every box, then add along the diagonals from bottom right.

c. The answer is _____

## 2 Now the standard way.

a. Work out 27 × 34 the way you were taught in class.

_____

b. Same answer? _____

c. Count the single-digit multiplications you did. _____ Compare with the lattice. _____

## 3 Predict before you do it.

| Problem | boxes in the lattice | single-digit facts |
|---|---|---|
| 46 × 23 | | |
| 7 × 385 | | |
| 124 × 56 | | |

**✎ ADD TO YOUR PROOF FILE**

Proof File page 29: both methods for 27 × 34 side by side, with the count of times-table facts under each. Same count, different bookkeeping.

**💬 TALK ABOUT IT**

Neither method does fewer times-table facts than the other. So what is actually different between them?

Which one hides the carrying, and is hiding it a good thing?



**STUCK? TRY THIS**

Count the boxes before you fill any in. A two-digit number times a two-digit number always needs 2 × 2 boxes, and that count does not depend on the method.

**★ CHALLENGE ZONE**

Work out 385 × 47 with the lattice, and check it by casting out nines from Week 16.

How many single-digit facts for a three-digit number times a three-digit number? Write the rule for a digits by b digits.

WEEK 32 STEPS & RULES

# The Euclid Machine

Take the smaller from the bigger, over and over. It always stops, and where it stops matters.

**IN CLASS THIS WEEK**
Multi-step problems; factors (4.OA.3, 4.OA.4)

**NUMBER WARM-UP · 2 MINUTES**
48 − 18 = ? Now take 18 away from your answer.

**WORDS TO KEEP**

**the machine** take the smaller number from the bigger, and repeat until the two are equal

**terminates** a method that is guaranteed to stop, rather than run forever

Run the machine on **48** and **18**. Every row takes the smaller away from the bigger.

| bigger | smaller | take it away |
|---|---|---|
| 48 | 18 | 48 − 18 = 30 |
| 30 | 18 | 30 − 18 = 12 |
| 18 | 12 | 18 − 12 = 6 |
| 12 | 6 | 12 − 6 = 6 |

**both are now 6, so the greatest common factor is 6**

**1 Check it against Week 7.**

a. The machine stopped at _____

b. Work out the greatest common factor of 48 and 18 by listing factors, the Week 7 way.

_____

c. Same answer? _____ Which method was less work? _____

## 2 Run it three more times.

| Start with | subtractions | stops at |
|---|---|---|
| 35 and 14 | | |
| 30 and 12 | | |
| 21 and 13 | | |

One pair stopped at 1. What does that say about those two numbers?

## 3 Why it cannot run forever.

Every row makes one of the two numbers **smaller**, and neither can go below 1. Finish the argument for why the machine always stops.

**✎ ADD TO YOUR PROOF FILE**

Proof File page 30: the machine's steps for 48 and 18, and the reason it must stop. That reason is a cousin of Week 1. There you watched a number that could never change; here you watch a number that can only go down.

**💬 TALK ABOUT IT**

The machine never looks for factors and never divides. How does it find the greatest common factor anyway?

Which pair took the most subtractions? What was unusual about it?



**STUCK? TRY THIS**

Write the pair on each line and subtract. When the two numbers finally match, that matching number is the answer.

**★ CHALLENGE ZONE**

Run the machine on 1,071 and 462. It is faster than you expect.

Find the pair of numbers, both from 2 to 29, that takes the **most** subtractions. It is not the pair you would guess, and once you see it the reason is obvious.

WEEK 33 FRACTIONS

# The Same Point on the Line

Equivalent fractions are not two things that are equal. They are one point with two names.

**IN CLASS THIS WEEK**
Equivalent fractions (4.NF.1, 4.NF.2)

**NUMBER WARM-UP · 2 MINUTES**
How many fourths make one half? How many eighths?

**WORDS TO KEEP**

**equivalent** different names for the same amount

**the same point** on one number line, equivalent fractions land on exactly the same mark

One line from 0 to 1, ticked three ways. Read straight **down** a tick to find every name for that point.




**1 Read down the ticks.**

a. The halfway mark is called _____ halves, _____ fourths and _____ eighths.

b. Three fourths is also _____ eighths.

c. Is there a mark for one third on this line? _____ Why not?

_____

## 2 Multiply top and bottom.

Cutting every piece into 2 doubles the number of pieces **and** the number you take. The point does not move.

| Fraction | × 2 top and bottom | × 3 top and bottom |
|---|---|---|
| 1 over 2 | 2 over 4 | |
| 3 over 4 | | |
| 2 over 3 | | |
| 5 over 6 | | |

## 3 Compare two that do not share a bottom.

Which is bigger, **2 over 3** or **3 over 4**? Rename both with a bottom of 12 first.

_____ and _____, so the bigger one is _____

✎ **ADD TO YOUR PROOF FILE**

Proof File page 31: the number line with the shared ticks marked, and the rule that multiplying top and bottom by the same number never moves the point.

💬 **TALK ABOUT IT**

Why does multiplying the top and bottom by the same number leave the amount alone?

Two fractions land on different points. Can renaming them ever make them equal?



**STUCK? TRY THIS**

To compare two fractions, rename them both so they have the **same bottom number**. Then the bigger top wins, and it is safe.

★ **CHALLENGE ZONE**

Find three fractions equivalent to 3 over 5, and mark them all on one number line.

Put these in order without a calculator: 2 over 3, 5 over 8, 3 over 4. Pick a bottom number that works for all three.

WEEK 34 FRACTIONS

# Parts of a Hundred

A hundred squares, and two ways to say the same shading.

**IN CLASS THIS WEEK**
Decimal notation for fractions (4.NF.5, 4.NF.6)

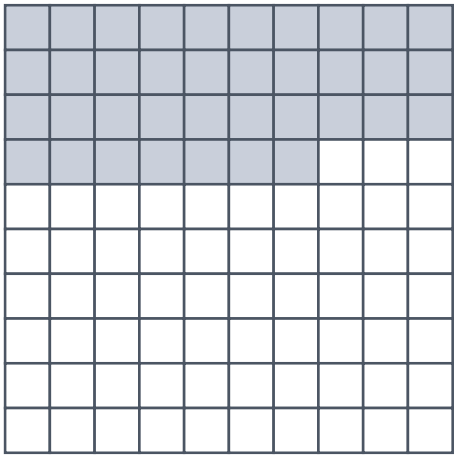
**NUMBER WARM-UP · 2 MINUTES**
How many hundredths in one tenth? In one half?

**WORDS TO KEEP**

**hundredth** one square of a hundred

**decimal point** everything after it is parts of one, not whole ones

Thirty-seven of the hundred squares are shaded.



**37 out of 100**

as a fraction _____

as a decimal _____

## 1 Two names, one picture.

a. As a fraction of a hundred: _____

b. As a decimal: _____

c. How many **whole rows** are shaded, and how many spare squares? _____ and _____.
So it is _____ tenths and _____ hundredths.

## 2 Fill in the table.

| Shaded squares | fraction of 100 | tenths and hundredths | decimal |
|---|---|---|---|
| 37 | 37 over 100 | 3 and 7 | 0.37 |
| 50 | | | |
| 8 | | | |
| 70 | | | |
| 25 | | | |

## 3 The awkward one.

a. Shade one quarter of the grid. How many squares? _____ As a decimal: _____

b. Now try one **eighth**. How many squares would that be? _____ Can you shade that many whole squares? _____

✎ **ADD TO YOUR PROOF FILE**

Proof File page 32: the shaded grid with all three names written beside it, and the eighth problem, which is the one worth remembering.

💬 **TALK ABOUT IT**

Why is 0.7 the same as 0.70 but not the same as 0.07?

Which fractions can be written exactly with a hundred squares, and which cannot?



**STUCK? TRY THIS**

Count whole rows first. Each whole row is one tenth, and each spare square is one hundredth, which is exactly what the two decimal places mean.

★ **CHALLENGE ZONE**

Which of these can be shaded exactly on a hundred grid: one half, one third, one fifth, one eighth, one twentieth?

What is special about the bottom numbers that work? Compare their prime factors with the prime factors of 100.

WEEK 35 LOGIC & PROOF

# The Weighing Puzzles

How much can one weighing possibly tell you? Count before you weigh.

**IN CLASS THIS WEEK**
Multi-step problems and reasoning (4.OA.3)

**NUMBER WARM-UP · 2 MINUTES**
3 × 3 = ?  3 × 3 × 3 = ?

**WORDS TO KEEP**

**outcome** one of the things that can happen. A balance has three: left down, right down, or level

**information** how many different cases a set of weighings could possibly tell apart

Nine coins. One is heavier than the rest and they all look the same. You have a balance and **two** weighings.




**coins 7, 8 and 9 stay off**

## 1 Count what a weighing can tell you.

a. How many things can happen when you weigh? _____ Name them.

_____

b. So two weighings can tell apart at most 3 × 3 = _____ different cases.

## 2 Now do it.

a. The picture shows the first weighing. If the left side goes down, which coins is the heavy one among? _____

b. If it balances? _____

c. Either way you are down to _____ coins. Describe the second weighing.

_____

## 3 How many coins could you handle?

| Weighings | cases you can tell apart | most coins |
|---|---|---|
| 1 | 3 | |
| 2 | 9 | |
| 3 | | |
| 4 | | |

**✎ ADD TO YOUR PROOF FILE**

Proof File page 33: the three-outcome count and the table. This is the last new page. The count tells you what is **impossible**, and the three-group method shows what is **possible**.

**💬 TALK ABOUT IT**

Twelve coins and two weighings. Say why it cannot be done, before trying anything.

The counting says three weighings could handle 27 coins. Does that mean a method exists?



**STUCK? TRY THIS**

Split into **three** equal groups, not two. A balance has three outcomes, so three groups is what uses a weighing fully.

**★ CHALLENGE ZONE**

Twenty-seven coins, one heavy, three weighings. Describe your first weighing.

Now the hard version: twelve coins, one is the odd weight but you do not know whether it is heavy or light. How many cases is that? Compare with what three weighings can tell apart.

WEEK 36 LOGIC & PROOF

# The Proof File, Finished

Thirty-three pages of reasons. This week they become one folder.

**IN CLASS THIS WEEK**
Review of the year

**NUMBER WARM-UP · 2 MINUTES**
No warm-up. Get every Proof File page out and put them in order.

Check your file against this list. Every page should have a **claim** and a **reason**, not just an answer.

## 1 The check list

| Weeks | The page should say | ✓ |
|---|---|---|
| 1 to 2 | invariants; parity rules things out | |
| 3 to 7 | factors, primes, trees, common factors and multiples | |
| 8 to 9 | coloring, and the cut chessboard | |
| 10 to 13 | degrees add to twice the lines; the tracing rule; three true claims; my tools so far | |
| 14 to 18 | divisibility with reasons; casting out nines; the wheel | |
| 19 to 21 | fold lines, turn order, the eight moves | |
| 22 to 25 | angles add to 180; which shapes tile; impossible tilings; my tools, weeks 14 to 24 | |
| 26 to 28 | choosing teams; the triangle; what a row total counts | |
| 29 to 32 | trees, cheapest networks, two multiplications, the machine | |
| 33 to 35 | one point two names; a hundred squares; counting outcomes | |

## 2 Four last claims.

Prove or disprove. Name the tool.

a. Eleven people can each shake hands with exactly three others. _____

b. 51 is prime. _____

c. A regular octagon tiles a floor on its own. _____

d. A tree with 20 dots has 19 lines. _____

## 3 Teach one page.

Pick the proof you are proudest of and give it out loud to somebody who has not seen it, without reading. Write down the first thing they did not believe.

_____

**✎ ADD TO YOUR PROOF FILE**

Write a first page for the file: your name, the year, and one sentence saying what a proof is, in your own words. Then number every page and make a contents list.

**💬 TALK ABOUT IT**

Which was harder this year: showing something is possible, or showing it is not?

Is there a page you can no longer explain? Go back to it before you call the file finished.



**STUCK? TRY THIS**

A page you cannot say out loud is not finished. That is the only test this week.

**★ CHALLENGE ZONE**

Look back at Week 1 and Week 29. The cups and the trees are both arguments about something that never changes. Write down what it is in each week.

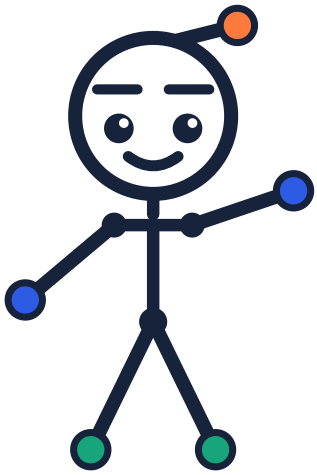
Invent one impossibility of your own, prove it, and put it on the last page of the file.

CERTIFICATE OF COMPLETION

# Prover of Impossible Things

awarded to

for a whole year of finding what a move cannot change, coloring a board to settle it, counting degrees, folding and turning shapes, running the machine, and, hardest of all, proving that something *cannot be done at all*.



**a Young Innovator for Change**

Teacher

Date

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